

# Lab09 – Deployment Horizontal Autoscaler

1. Install the metrics server (<https://raw.githubusercontent.com/brahimhamdi/ckad-lab/master/labs/lab09/components.yaml>) to actual resource metrics to be collected and displayed by the HorizontalPodAutoscaler.

```
vagrant@kubernetes-control-plane:~/lab$ kubectl apply -f https://raw.githubusercontent.com/brahimhamdi/ckad-lab/master/labs/lab09/components.yaml
serviceaccount/metrics-server created
clusterrole.rbac.authorization.k8s.io/system:aggregated-metrics-reader created
clusterrole.rbac.authorization.k8s.io/system:metrics-server created
rolebinding.rbac.authorization.k8s.io/metrics-server-auth-reader created
clusterrolebinding.rbac.authorization.k8s.io/metrics-server:system:auth-delegator created
clusterrolebinding.rbac.authorization.k8s.io/system:metrics-server created
service/metrics-server created
deployment.apps/metrics-server created
apiservice.apiregistration.k8s.io/v1beta1.metrics.k8s.io created
vagrant@kubernetes-control-plane:~/lab$ kubectl get pod -A
```

| NAMESPACE   | NAME                                       | READY | STATUS  | RESTARTS      | AGE   |
|-------------|--------------------------------------------|-------|---------|---------------|-------|
| kube-system | coredns-5d78c9869d-5gl97                   | 1/1   | Running | 0             | 6h4m  |
| kube-system | coredns-5d78c9869d-dnts9                   | 1/1   | Running | 0             | 6h4m  |
| kube-system | etcd-kube-control-plane                    | 1/1   | Running | 0             | 6h4m  |
| kube-system | kube-apiserver-kube-control-plane          | 1/1   | Running | 0             | 6h4m  |
| kube-system | kube-controller-manager-kube-control-plane | 1/1   | Running | 0             | 6h4m  |
| kube-system | kube-proxy-24w7b                           | 1/1   | Running | 0             | 5h51m |
| kube-system | kube-proxy-kgm9f                           | 1/1   | Running | 0             | 6h4m  |
| kube-system | kube-proxy-vxfm9                           | 1/1   | Running | 0             | 5h50m |
| kube-system | kube-scheduler-kube-control-plane          | 1/1   | Running | 0             | 6h4m  |
| kube-system | metrics-server-66758478b5-nns8n            | 1/1   | Running | 0             | 77s   |
| kube-system | weave-net-245pv                            | 2/2   | Running | 0             | 5h50m |
| kube-system | weave-net-j4zh7                            | 2/2   | Running | 1 (5h56m ago) | 5h57m |
| kube-system | weave-net-nff2s                            | 2/2   | Running | 0             | 5h51m |

```
vagrant@kubernetes-control-plane:~/lab$
```

2. Create a Deployment named `nginx` with 1 replica. The Pod template of the Deployment should use container image `nginx:1.23.4`, set the CPU resource request to 0.1, and the memory resource request/limit to 100Mi.

```
vagrant@kubernetes-control-plane:~/lab$ vim nginx-deployment.yaml
vagrant@kubernetes-control-plane:~/lab$ cat nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx
spec:
  replicas: 1
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - image: nginx:1.23.4
          name: nginx
          resources:
            requests:
              cpu: "0.1"
              memory: "100Mi"
            limits:
              memory: "200Mi"
vagrant@kubernetes-control-plane:~/lab$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx created
vagrant@kubernetes-control-plane:~/lab$ kubectl get deploy,pod
```

| NAME                  | READY | UP-TO-DATE | AVAILABLE | AGE |
|-----------------------|-------|------------|-----------|-----|
| deployment.apps/nginx | 1/1   | 1          | 1         | 29s |

```
NAME                                READY    STATUS    RESTARTS    AGE
pod/nginx-554c9d466c-trgc2         1/1     Running   0           29s
vagrant@kubernetes-control-plane:~/lab$
```

3. Create a HorizontalPodAutoscaler for the Deployment named `nginx-hpa` that scales to minimum of 2 and a maximum of 5 replicas. Scaling should happen based on an average CPU utilization of 50%, and an average memory utilization of 60%.

```
vagrant@kubernetes-control-plane:~/lab$ vim nginx-hpa.yaml
vagrant@kubernetes-control-plane:~/lab$ cat nginx-hpa.yaml
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: nginx-hpa
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: nginx
  minReplicas: 2
  maxReplicas: 5
  metrics:
  - type: Resource
    resource:
      name: cpu
      target:
        type: Utilization
        averageUtilization: 50
  - type: Resource
    resource:
      name: memory
      target:
        type: Utilization
        averageUtilization: 60
vagrant@kubernetes-control-plane:~/lab$ kubectl apply -f nginx-hpa.yaml
horizontalpodautoscaler.autoscaling/nginx-hpa created
vagrant@kubernetes-control-plane:~/lab$
```

4. Inspect the HorizontalPodAutoscaler object and identify the currently-utilized resources. How many replicas do you expect to exist?

```
vagrant@kubernetes-control-plane:~/lab$ kubectl get hpa nginx-hpa
NAME          REFERENCE          TARGETS          MINPODS  MAXPODS  REPLICAS  AGE
nginx-hpa     Deployment/nginx    0%/50%, 2%/60%   2         5         2         56s
vagrant@kubernetes-control-plane:~/lab$
```

5. Create a service to expose nginx deployment. Generate traffic with busybox image and then watch the HPA object.

```
kubectl run -i --tty load-generator --rm --image=busybox:1.28 --restart=Never -- /bin/sh -c
"while true; do wget -q -O- http://my-service; done"
```

```
vagrant@kubernetes-control-plane:~/lab$ kubectl expose deployment nginx --name=my-service --port=80
service/my-service exposed
vagrant@kubernetes-control-plane:~/lab$ kubectl run -i --tty load-generator --rm --image=busybox:1.28 --restart=Never -- /bin/sh -c "while sleep 0.0
1; do wget -q -O- http://my-service; done"

vagrant@kubernetes-control-plane:~/lab$ kubectl get hpa nginx-hpa
NAME          REFERENCE          TARGETS          MINPODS  MAXPODS  REPLICAS  AGE
nginx-hpa     Deployment/nginx    50%/50%, 2%/60%   2         5         2         12m
vagrant@kubernetes-control-plane:~/lab$
```